

^{35}Ar $\varepsilon+\beta^+$ decay (1.7755 s) 1980Wi13,1984Ad01

Parent: ^{35}Ar : $E=0$; $J^\pi=3/2^+$; $T_{1/2}=1.7755$ s 14; $Q(\varepsilon)=5966.2$ 7; $\% \varepsilon+\% \beta^+$ decay=100

^{35}Ar - J^π , $T_{1/2}$: From Adopted Levels of ^{35}Ar .

^{35}Ar - $Q(\varepsilon)$: From 2021Wa16.

1980Wi13: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 11.0-MeV protons from the ONR-CIT tandem accelerator impinging on LiCl and PbCl_2 wrapped in 25- μm Ta foil. γ rays were detected using a 100- cm^3 $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for 20 γ -ray peaks. Deduced decay branching ratios for 9 levels, and $\log ft$. Compared with shell-model calculations.

1984Ad01: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 13-MeV protons from the University of Washington FN tandem accelerator impinging on a thick KCl target. γ rays were detected using a 28% $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for 8 γ -ray peaks. Deduced decay branching ratios for 6 levels, and $\log ft$. Compared with shell-model calculations.

1985Da04: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 9-MeV protons from the Van de Graaff accelerator at the University of Pittsburgh impinging on a NaCl target. γ rays were detected using a $\text{Ge}(\text{Li})$ detector with NaI anti-Compton shielding. Measured E_γ and I_γ for 4 γ -ray peaks. Deduced 3 levels.

1979Ha04: ^{35}Ar produced via the $^{51}\text{V}(p,p11n)$ spallation reaction with 600-MeV protons from the CERN synchrocyclotron bombarding a 38- g/cm^2 vanadium target in the form of VC powder. γ rays were detected using a $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for the 1219 γ . Deduced the decay branching ratio for the 1219-keV level.

1977Az01: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 10-13-MeV protons from the McGill synchrocyclotron impinging on CCl_4 and LiCl targets and 8-MeV protons from the University of Montreal impinging a thick LiCl pellet. Positrons were detected using an NE102 plastic scintillator coupled to an RCA 6342A photomultiplier tube. γ rays were detected using an 18% $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for 3 γ -ray peaks. Deduced decay branching ratios for 3 levels and parent ^{35}Ar $T_{1/2}$. Related: $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction threshold measurement in 1978Az01.

1971Ge04: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 10-MeV protons from the Chalk River tandem accelerator impinging on PbCl_2 targets. γ rays were detected using two coaxial $\text{Ge}(\text{Li})$ detectors. Measured E_γ and I_γ for 4 γ -ray peaks. Deduced 4 levels, decay branching ratios, $\log ft$, and parent ^{35}Ar $T_{1/2}$.

1971De05: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction on KCl targets. γ rays were detected using a 25 cm^3 $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ . Deduced 6 levels, decay branching ratios, and $\log ft$.

1969Wi18,1969Fr09: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with a proton beam from the Harwell tandem accelerator impinging on 3.75- mg/cm^2 KCl deposited on 0.13-mm Au backings and 160-mmHg gas CCl_4 targets. γ rays were detected using a $\text{Ge}(\text{Li})$ detector. Measured E_γ and I_γ for 2 γ -ray peaks. Deduced 2 levels, decay branching ratios, (p,n) reaction threshold and β end-point energy, $\log ft$, and parent ^{35}Ar $T_{1/2}$.

1960Wa04: ^{35}Ar was produced at Crocker Laboratory and the Lawrence Radiation Laboratory. β spectra were measured with a uniform-field 180°-deflection single-focusing spectrometer. Deduced Coulomb-energy differences between mirror ground-state transitions.

1956Ki29: ^{35}Ar was produced via the $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction with 10-MeV protons from the BNL 60-inch cyclotron impinging on a liquid CCl_4 target. β particles were detected using a conventional thin-lens, iron-free, magnetic spectrometer. γ rays were detected using a $\text{NaI}(\text{Ti})$ crystal scintillation counter. Measured E_β and total I_β for two positron decays to ^{35}Cl excited states, E_γ and I_γ for two γ -ray peaks. Deduced two levels and decay branching ratios.

^{35}Ar β asymmetry parameter measurements: 1993Co07, 1991Na05, 1991Mi18, 1988Ga02, 1965Ca04.

 ^{35}Cl Levels

$E(\text{level})^\dagger$	$J^\pi^\ddagger$	$T_{1/2}^\ddagger$
0	$3/2^+$	stable
1219.2 2	$1/2^+$	0.119 ps 14
1763.1 2	$5/2^+$	0.36 ps 3
2645.1 10	$7/2^+$	0.159 ps 21
2693.9 2	$3/2^+$	32 fs 6
3002.2 4	$5/2^+$	22 fs 6
3918.3 15	$3/2^+$	4.9 fs 14
3967.9 6	$1/2^+$	11.8 fs 28
4624.5? 5	$3/2, 5/2$	40 fs 17

† From a least-squares fit to γ -ray energies.

‡ From the Adopted Levels.

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ε, β^+ radiations						Comments
E(decay)	E(level)	$I\beta^+$ ‡	$I\varepsilon$ ‡	Log ft	$I(\varepsilon+\beta^+)$ ‡‡	
(1341.7 $^{\#}$ 13)	4624.5?	$<8.0\times 10^{-4}$	<0.0035	>3.8	<0.0043	$I(\varepsilon+\beta^+)$: deduced from $100-\Sigma I(\varepsilon+\beta^+)$ (excited states), where $\Sigma I(\varepsilon+\beta^+)$ (excited states)=1.76 5. Level scheme fed by β^+ decay may be complete considering the highest observed level is at 4.6 MeV. Observed $\Sigma I(\varepsilon)\approx 0.08$. Evaluators assume $\Sigma I(\varepsilon)\leq 0.08$ for all unobserved ε feedings to high-lying levels and include this limit in the uncertainty for $\Sigma I(\varepsilon+\beta^+)$ (excited states)=1.76 10 and $I(\varepsilon+\beta^+)$ (gs)=98.24 10.
(1998.3 14)	3967.9	0.015 10	0.0016 10	$4.5+5-2$	0.017 11	
(2047.9 19)	3918.3	0.0096 30	8.3×10^{-4} 26	$4.8+2-1$	0.0104 32	
(2964.0 13)	3002.2	0.089 14	0.0011 2	5.0 1	0.090 14	
(3272.3 13)	2693.9	0.164 9	0.00126 7	5.00 3	0.165 9	
(3321.1 16)	2645.1	<0.00146	$<2.82\times 10^{-5}$	>8.3	<0.00149	
(4203.1 13)	1763.1	0.256 13	7.0×10^{-4} 4	5.47 2	0.257 13	
(4747.0 13)	1219.2	1.22 4	0.00208 7	5.10 2	1.22 4	
(5966.2 16)	0	98.17 10	0.07309 8	3.759 1	98.24 10	

‡ Deduced from γ -ray intensity balance, relative to $I\gamma(1219\gamma)=100$, normalized to absolute $\%I\gamma(1219\gamma)=1.26$ 5.

‡‡ Absolute intensity per 100 decays.

$^{\#}$ Existence of this branch is questionable.

³⁵Ar ε+β⁺ decay (1.7755 s) [1980Wi13,1984Ad01](#) (continued)

γ(³⁵Cl)

I_γ normalization: absolute %I_γ(1219γ)=1.243 30 deduced from weighted average of 1.223 46 ([1969Wi18,1969Fr09](#)), 1.22 20 ([1971Ge04](#)), 1.55 15 ([1977Az01](#)), 1.34 8 ([1979Ha04](#)), 1.23 7 ([1984Ad01](#)), 1.228 30 ([1980Wi13](#)).

E _γ [†]	I _γ ^{‡&}	E _i (level)	J _i ^π	E _f	J _f ^π	Mult. [#]	δ [#]	α ^a	Comments
916 ^b		3918.3	3/2 ⁺	3002.2	5/2 ⁺				The intensity could not be determined due to being obscured by another peak (1980Wi13).
930.7 2	1.57 20	2693.9	3/2 ⁺	1763.1	5/2 ⁺	M1+E2	+0.09 3	4.69×10 ⁻⁵ 7	%I _γ =0.0195 25 I _γ : weighted average of 1.13 26 (1980Wi13) and 1.66 12 (1984Ad01).
965 ^b	<0.87 [@]	3967.9	1/2 ⁺	3002.2	5/2 ⁺				%I _γ <0.0111
1219.3 2	100.0 15	1219.2	1/2 ⁺	0	3/2 ⁺	M1+E2	0.093 10	3.66×10 ⁻⁵ 5	%I _γ =1.24 4 I _γ : others: 100.0 30 from 1984Ad01 , 100 10 from 1977Az01 , 100 from 1969Wi18 , 1971Ge04 , 1979Ha04 , and 1985Da04 .
1225 ^b		3918.3	3/2 ⁺	2693.9	3/2 ⁺				The intensity could not be determined due to being obscured by another peak (1980Wi13).
1239 ^b	<0.30 [@]	3002.2	5/2 ⁺	1763.1	5/2 ⁺	[M1,E2]		4.4×10 ⁻⁵ 6	%I _γ <0.00382
1274 ^b	<0.42 [@]	3967.9	1/2 ⁺	2693.9	3/2 ⁺	[M1,E2]		4.9×10 ⁻⁵ 7	%I _γ <0.00535
1474.8 3	1.02 11	2693.9	3/2 ⁺	1219.2	1/2 ⁺	M1+E2	-0.63 21	8.4×10 ⁻⁵ 4	%I _γ =0.0127 14 I _γ : weighted average of 0.83 13 (1980Wi13) and 1.08 7 (1984Ad01).
1763.0 2	22.7 4	1763.1	5/2 ⁺	0	3/2 ⁺	M1+E2	-2.85 10	0.0002124 30	%I _γ =0.282 9 I _γ : weighted average of 23.4 13 (1969Wi18), 20.0 27 (1971Ge04), 22 6 (1977Az01), 23.08 37 (1980Wi13), 21.85 67 (1984Ad01), and 21.7 9 (1985Da04).
1783 ^b	<0.39 [@]	3002.2	5/2 ⁺	1219.2	1/2 ⁺	[E2]		0.0002267 32	%I _γ <0.00497
1931 ^b	<0.21 [@]	4624.5?	3/2,5/2	2693.9	3/2 ⁺				%I _γ <0.00267
2155.1 15	0.15 6	3918.3	3/2 ⁺	1763.1	5/2 ⁺	[M1,E2]		0.00036 4	%I _γ =0.0019 8 I _γ : from 1984Ad01 . Other: <0.22 (1980Wi13).
2204 ^b	<0.27 [@]	3967.9	1/2 ⁺	1763.1	5/2 ⁺	[E2]		0.000425 6	%I _γ <0.00344
2645 ^b	<0.12 [@]	2645.1	7/2 ⁺	0	3/2 ⁺	E2		0.000633 9	%I _γ <0.00153
2693.7 2	10.96 18	2693.9	3/2 ⁺	0	3/2 ⁺	M1+E2	+0.26 2	0.000553 8	%I _γ =0.136 4 I _γ : weighted average of 10.2 15 (1971Ge04), 10.96 18 (1980Wi13) and 11.02 34 (1984Ad01).
2699 ^b	<0.33 [@]	3918.3	3/2 ⁺	1219.2	1/2 ⁺	[M1,E2]		0.00060 6	%I _γ <0.00420
2748.5 6	0.477 34	3967.9	1/2 ⁺	1219.2	1/2 ⁺	[M1]		0.000569 8	%I _γ =0.0059 5 I _γ : weighted average of 0.48 6 (1980Wi13) and 0.476 34 (1984Ad01).

³⁵Ar $\varepsilon+\beta^+$ decay (1.7755 s) [1980Wi13](#),[1984Ad01](#) (continued)

$\gamma(^{35}\text{Cl})$ (continued)

E_γ [†]	I_γ ^{‡&}	$E_i(\text{level})$	J_i^π	E_f	J_f^π	Mult. [#]	δ [#]	α ^a	Comments
2861 ^b	<0.099 [@]	4624.5?	3/2,5/2	1763.1	5/2 ⁺				%I _γ <0.00126
3002.1 4	7.29 14	3002.2	5/2 ⁺	0	3/2 ⁺	M1+E2	+0.08 2	0.000671 9	%I _γ =0.0906 28 I _γ : weighted average of 9.0 15 (1971Ge04), 7.24 14 (1980Wi13), 7.45 23 (1984Ad01) and 6.9 5 (1985Da04).
3405 ^b		4624.5?	3/2,5/2	1219.2	1/2 ⁺				I _γ : could not be determined due to being obscured by another peak (1980Wi13).
3918	0.52 3	3918.3	3/2 ⁺	0	3/2 ⁺	M1+E2		0.00108 7	%I _γ =0.0065 4
3968	0.106 25	3967.9	1/2 ⁺	0	3/2 ⁺	[M1,E2]		0.00109 7	%I _γ =0.00132 31
4624 ^b	<0.035 [@]	4624.5?	3/2,5/2	0	3/2 ⁺	D+Q			%I _γ <4.46×10 ⁻⁴

[†] From [1984Ad01](#), unless otherwise noted.

[‡] from [1980Wi13](#), unless otherwise noted.

[#] From Adopted Gammas.

[@] [1980Wi13](#) gives upper limits of I_γ at 98% confidence level. These peaks are not observed in the spectra of [1980Wi13](#) or in [1984Ad01](#).

[&] For absolute intensity per 100 decays, multiply by 0.01243 30.

^a Total theoretical internal conversion coefficients, calculated using the BrIcc code ([2008Ki07](#)) with “Frozen Orbitals” approximation based on γ -ray energies, assigned multipolarities, and mixing ratios, unless otherwise specified.

^b Placement of transition in the level scheme is uncertain.

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